Game Theory

Midterm Examination | Fall 2023

Prof. R. Alvarez

Time limit: 80 minutes • Total: 60 points

Name: _____ Student ID: _____

Instructions

- Closed book and closed notes. One single-sided, handwritten $8.5'' \times 11''$ sheet of notes is allowed.
- No calculators or electronic devices. Leave answers as exact fractions.
- Show your work. A correct final answer with no supporting reasoning receives little credit.
- In every bimatrix, the first entry is the payoff of the Row player and the second entry is the payoff of the Column player.
- "Nash equilibrium" (NE) includes mixed-strategy equilibria unless a question says "pure".

Problem	1	2	3	4	5	Total
Points	12	14	16	10	8	60
Score						

Problem 1 Iterated Dominance [12 points]

Consider the following game. Row chooses among U, M, D; Column chooses among a, b, c, d.

	a	b	c	d
U	4, 0	1, 3	2, 1	0, 2
M	0, 4	3, 2	2, 1	0, 1
D	1, 1	2, 3	1, 5	4, 2

(a) **[2]** Show that exactly one pure strategy (of either player) is strictly dominated by another pure strategy in the original game. Identify it.

(b) **[4]** After removing the strategy from part (a), show that D is *not* strictly dominated by U or by M, but that it *is* strictly dominated by the mixed strategy $\alpha U + (1-\alpha)M$ for suitable α. Find the set of all $\alpha \in [0,1]$ that work.

(c) **[2]** Complete the process of iterated elimination of strictly dominated strategies (IESDS). State the reduced game that survives.

(d) **[4]** Find all Nash equilibria of the *original* 3×4 game, and the expected payoff of each player in each equilibrium.

Problem 2 Pure and Mixed Equilibria in a 2×3 Game [14 points]

	L	C	R
T	3, 4	1, 3	2, 0
B	1, 0	4, 2	4, 3

(a) **[3]** Find all pure-strategy Nash equilibria.

(b) **[4]** Let p denote the probability that Row plays T. Write Column's expected payoff from each of L, C, R as a function of p, and give Column's best-response correspondence for every $p \in [0, 1]$.

(c) **[5]** Find all Nash equilibria in which at least one player uses a non-degenerate mixed strategy. Justify why there are no others.

(d) **[2]** List the expected payoff pair of every Nash equilibrium of the game. Is any equilibrium Pareto dominated by another equilibrium?

Problem 3 A Zero-Sum Game: the Graphical Method [16 points]

In the zero-sum game below, entries are payments from Column to Row (Row maximizes, Column minimizes).

	c_1	c_2	c_3	c_4
r_1	5	-2	2	0
r_2	-1	4	1	6

(a) **[2]** Show that the game has no saddle point in pure strategies, and identify a column that Column never needs to use.

(b) **[7]** Let Row play r_1 with probability p. Sketch Row's expected payoff against each column as a function of p, identify the lower envelope, and find Row's optimal (maximin) strategy p^* and the value v of the game.

(c) **[5]** Find Column's optimal strategy. Is it unique? Explain.

(d) **[2]** Suppose Row carelessly plays $p = \frac{1}{2}$. What is Column's best response, and what expected payoff does Row then receive?

Problem 4 Best-Response Correspondences [10 points]

	L	R
U	3, 1	0, 2
D	3, 2	1, 1

Let $p = \Pr(U)$ and $q = \Pr(L)$.

(a) **[3]** Derive Row's best-response correspondence $B_R(q)$ for all $q \in [0, 1]$.

(b) **[2]** Derive Column's best-response correspondence $B_C(p)$ for all $p \in [0, 1]$.

(c) **[4]** Sketch both correspondences in the unit square and find *all* Nash equilibria.

(d) **[1]** Which of these equilibria survive if we first delete weakly dominated strategies?

Problem 5 True or False [8 points]

Answer **True** or **False** and give a one- or two-sentence justification (a proof sketch or a counterexample). Each part is worth 2 points; no credit without justification.

(a) In a finite two-player zero-sum game, if (r, c) and (r', c') are both pure-strategy saddle points, then (r, c') is also a saddle point.

(b) A pure strategy that is weakly dominated can never be played with positive probability in a Nash equilibrium.

(c) In any Nash equilibrium of a finite game, every pure strategy that a player uses with positive probability gives that player the same expected payoff.

(d) A mixed strategy that assigns positive probability to a strictly dominated pure strategy can be a best response to some mixed strategy of the opponent.

— End of examination —